

Reading and writing to an SD card

The following instructions need to be kept in mind when reading and writing using the SD card shield library:

- ◆ On the Ethernet Shield, CS is Pin 4. It's set as an output by default.
- ◆ Note that even if it's not used as the CS pin, the hardware SS pin (10 on most Arduino™ boards, 53 on the Mega) must be left as an output or the SD library functions won't work.

You can find a copy of the code at <http://dgit.in/ardsdcode> and related explanation at <http://dgit.in/ardsdexp>.

Using Shields

To use a Shield with your Arduino™ boards, you need to place it on top of the Arduino™ board. You can also stack up multiple shields on top of each other

Using the Ethernet Shield

The Ethernet shield is very useful when you need to connect the Arduino™ board to many other devices or to the Internet. The board uses the SPI library to talk to the Ethernet shield.

If your application isn't performing network operations as needed and you suspect that the Ethernet shield is taking up the necessary processing cycles, it's better to use the boards mentioned in the Advanced Hardware section (Chapter 8).

You can find a copy of the code at <http://dgit.in/ardethscode> and related explanation at <http://dgit.in/ardethexp>.

Using the Wi-Fi Shield

The Wi-Fi shield uses the Serial Peripheral Interface (SPI) library to talk to the Wi-Fi chip and the SD card on board the Wi-Fi shield. Both libraries are available as part of standard Arduino™ installation.

#include <SPI.h>	Include the 'Serial Peripheral Interface' library
#include <WiFi.h>	Include the Arduino™ Wi-Fi library
char ssid[30];	Variable to store SSID (name) of your network
char pass[20];	Variable to store network password
int status = WL_IDLE_STATUS;	Set Wi-Fi wireless status to IDLE
void setup () {	
Serial.begin (9600);	Initialise serial port at baud rate of 9.6kbps
while (!Serial);	Wait for the port to open